

Assignments Continuous Integration Using Jenkins

Submitted to - Vikul Sir

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L3 - Create CI/CD Pipeline to Deploy the Maven Web Application in Tomcat Server

Jenkins + nexus + SonarQube + s3 + Tomcat

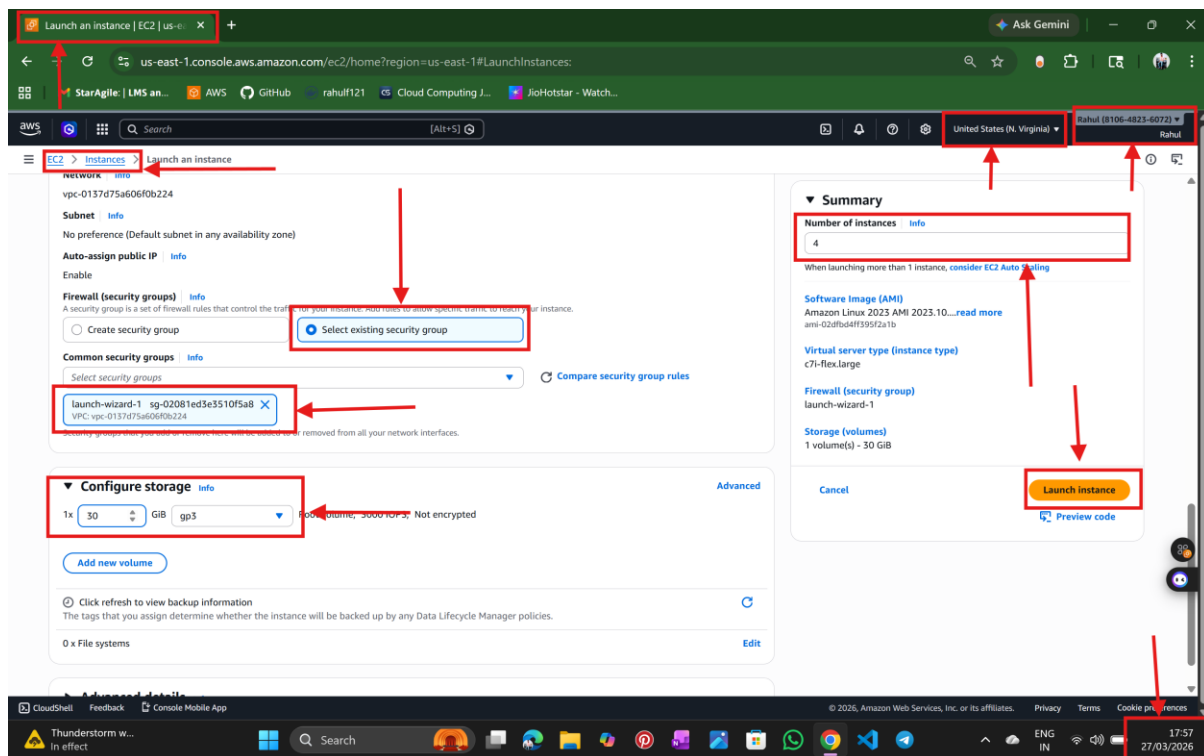
Step 1: launch a four ec2 instance

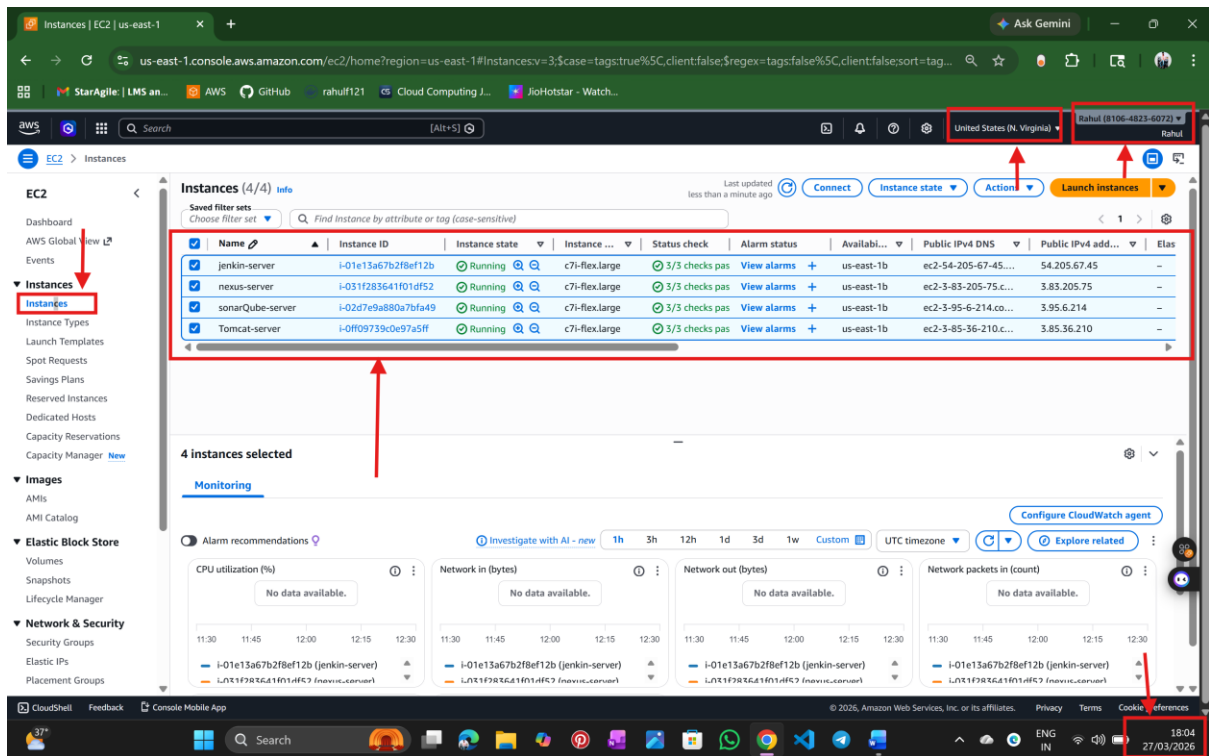
→ one for Jenkins

→ for tomcat

→ for nexus

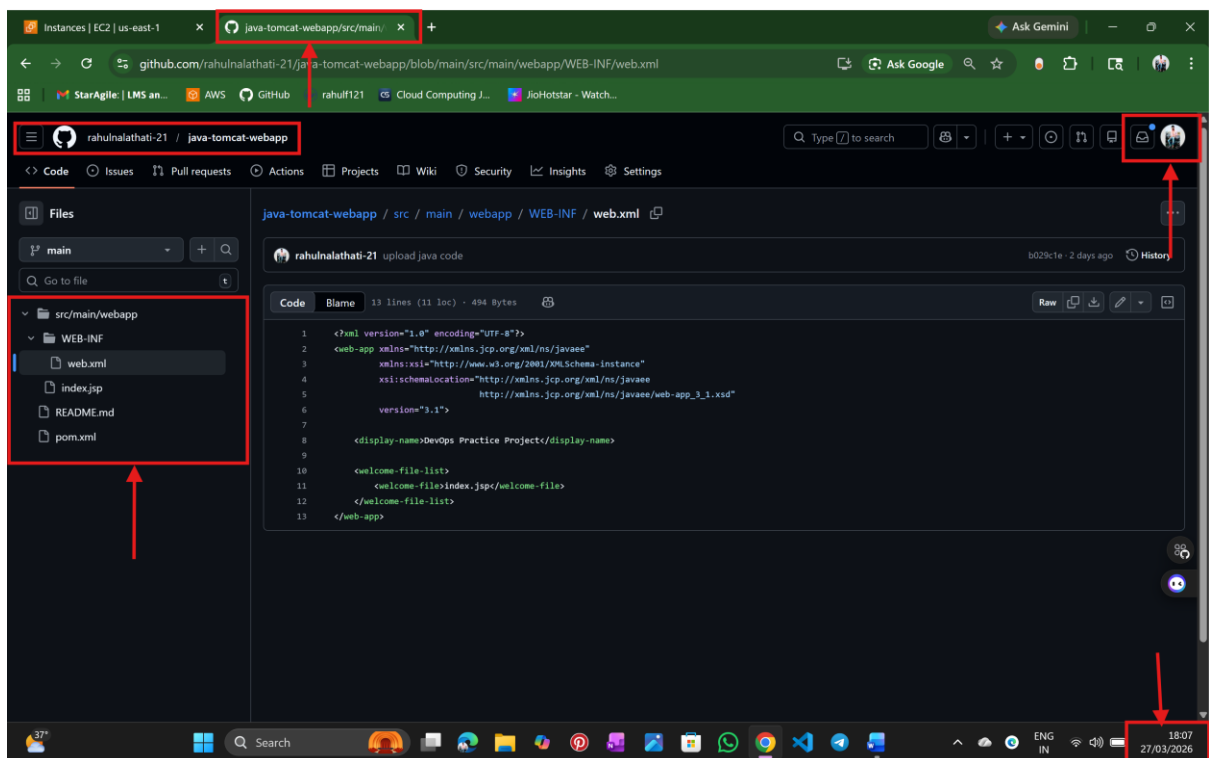
→ and for sonar Qube





Step 2: push source code of the web application to git hub

→ make sure, source code in git hub



Step 3: install Prerequisites

On Tomcat server

→ connect to mobXterm with public IP of the server

→ install maven

→ install tomcat

→ Configure Tomcat for Deployment

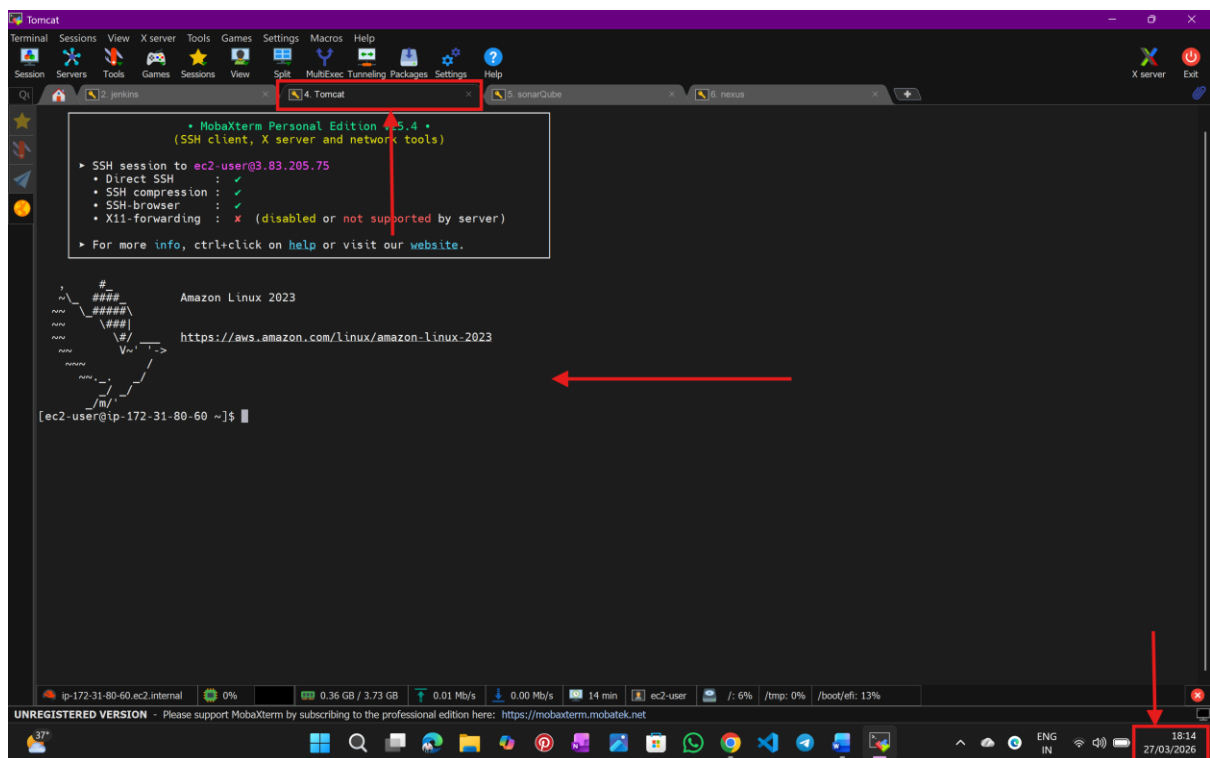
→ using public IP :<port number> access the Tomcat web server

Eg: public IP:8080 → by default Tomcat run on 8080 port number

→ To access the Tomcat Dashboard

Username: admin

Password: admin



Instances | EC2 | us-east-1 | Dashboard - Jenkins | /manager

Not secure 3.83.205.75:8080/manager/html

StarAgile | LMS an... | AWS | GitHub | rahulf121 | Cloud Computing J... | JioHotstar - Watch...

THE ASF

Tomcat Web Application Manager

Message: OK

Manager

[List Applications](#) [HTML Manager Help](#) [Manager Help](#) [Server Status](#)

Applications					
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

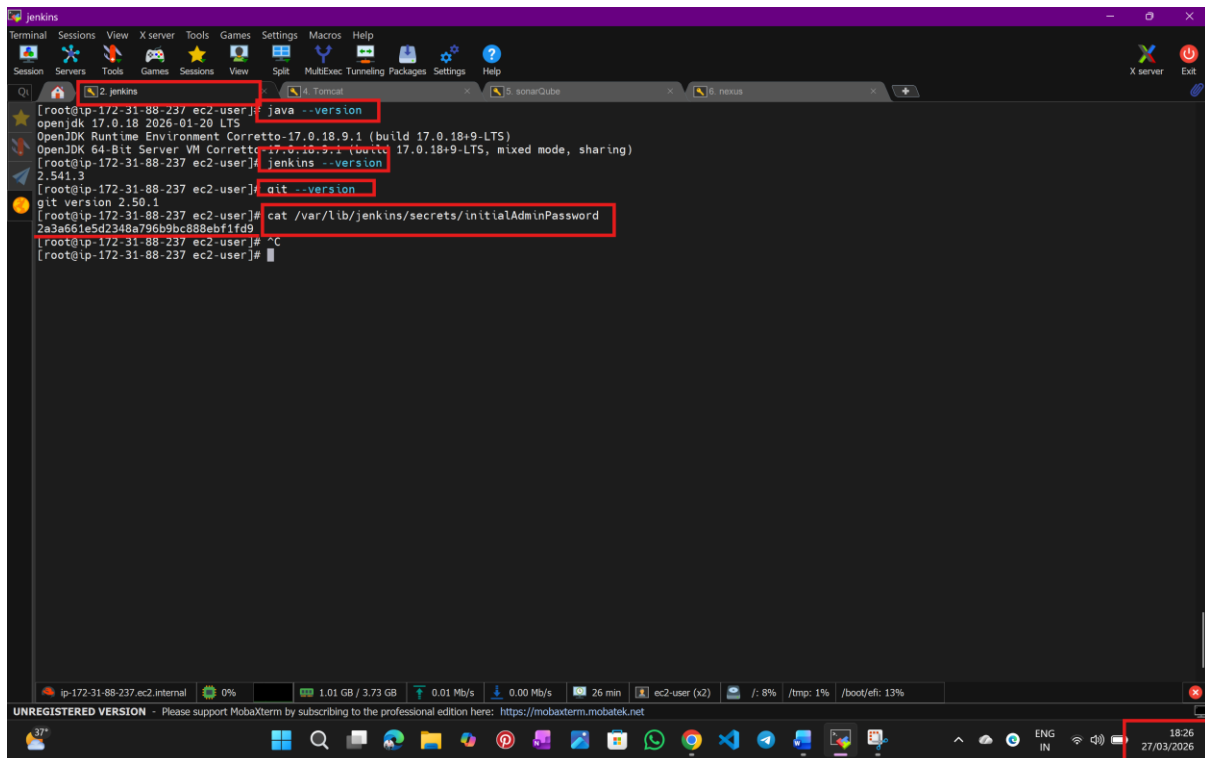
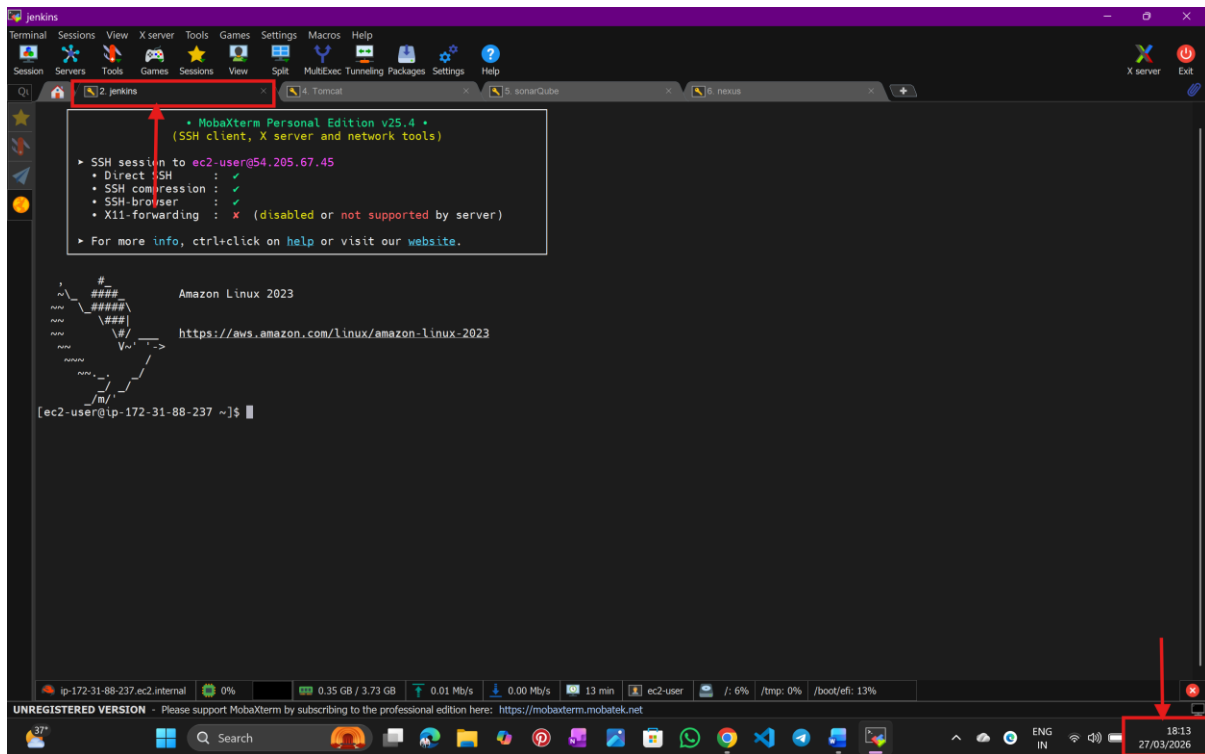
Deploy

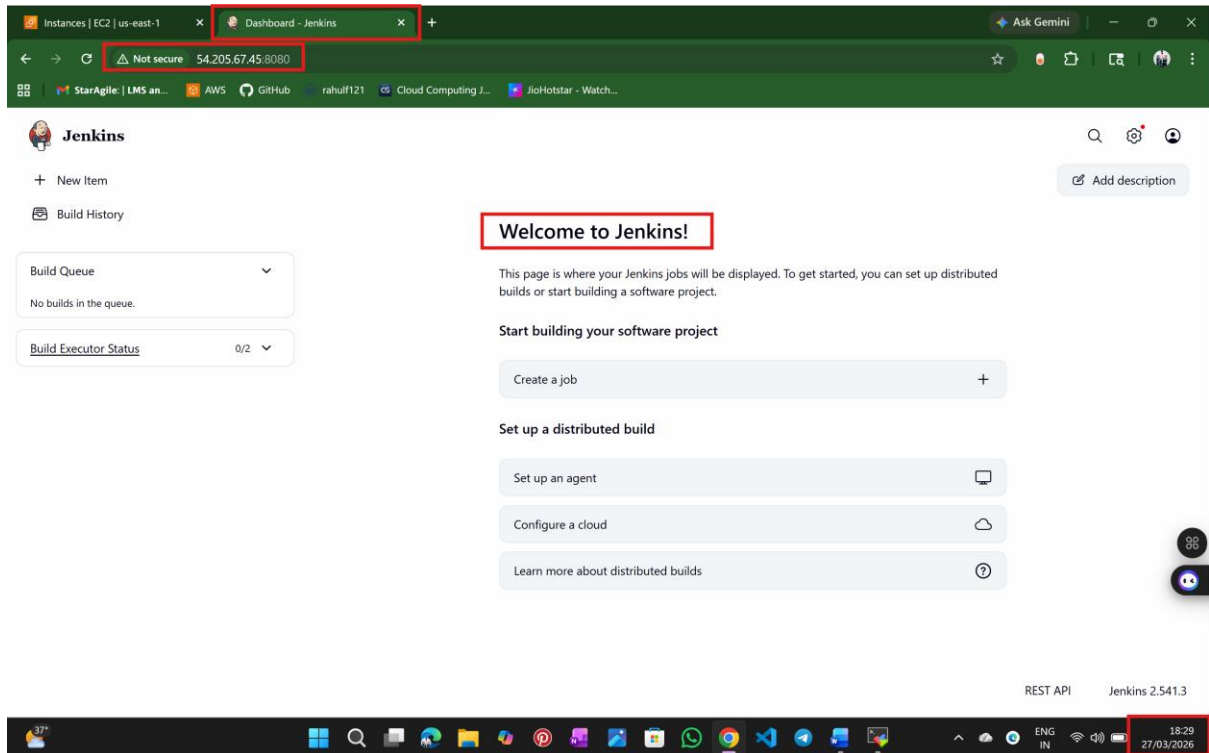
Deploy directory or WAR file located on server

18:55 27/03/2026

On Jenkins server

- connect to mobXterm with public IP of the server
- install java (JDK)
- install maven
- install Jenkins
- using public IP :<port number> access the Jenkins web server
- Eg: public IP:8080 → by default Jenkins run on 8080 port number
- To access the Jenkins Dashboard
- Username: <give>
- Password: <give>





On Nexus server

→ connect to mobXterm with public IP of the server

→ install docker

→ install maven

→ use the below command

`docker run -d -p <host port no>:<container port no> --name <container name> <docker image name>`

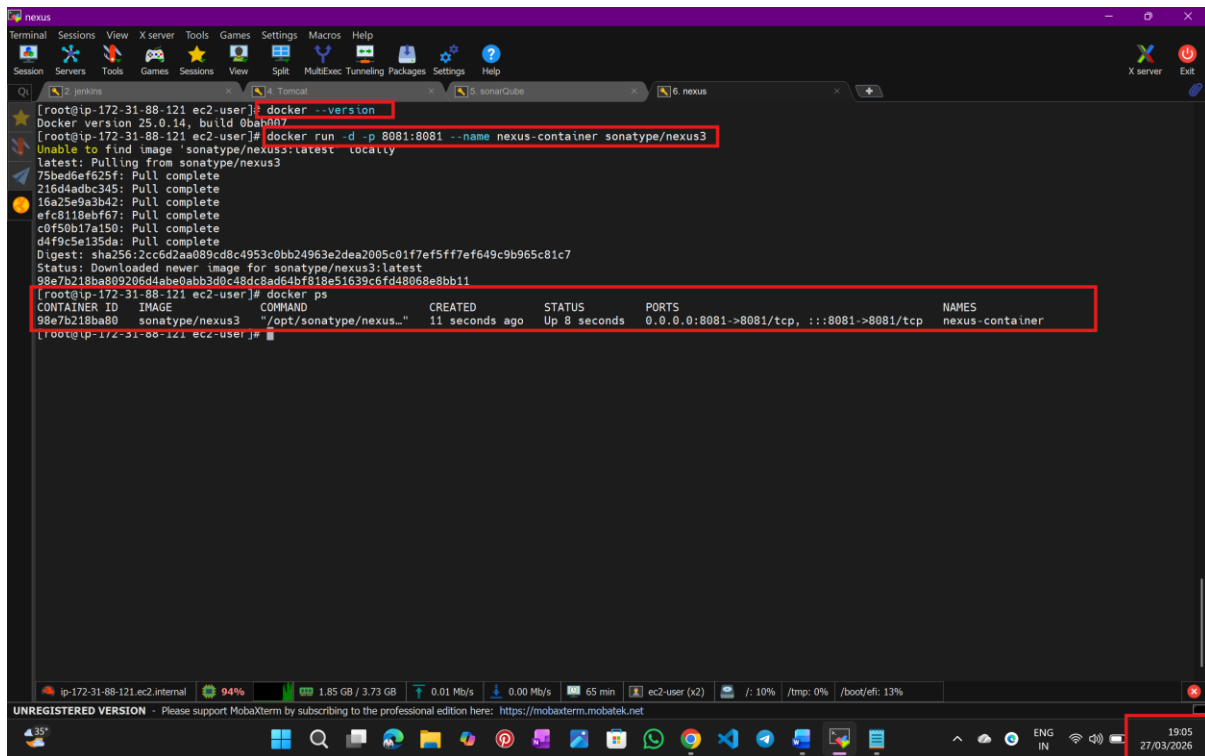
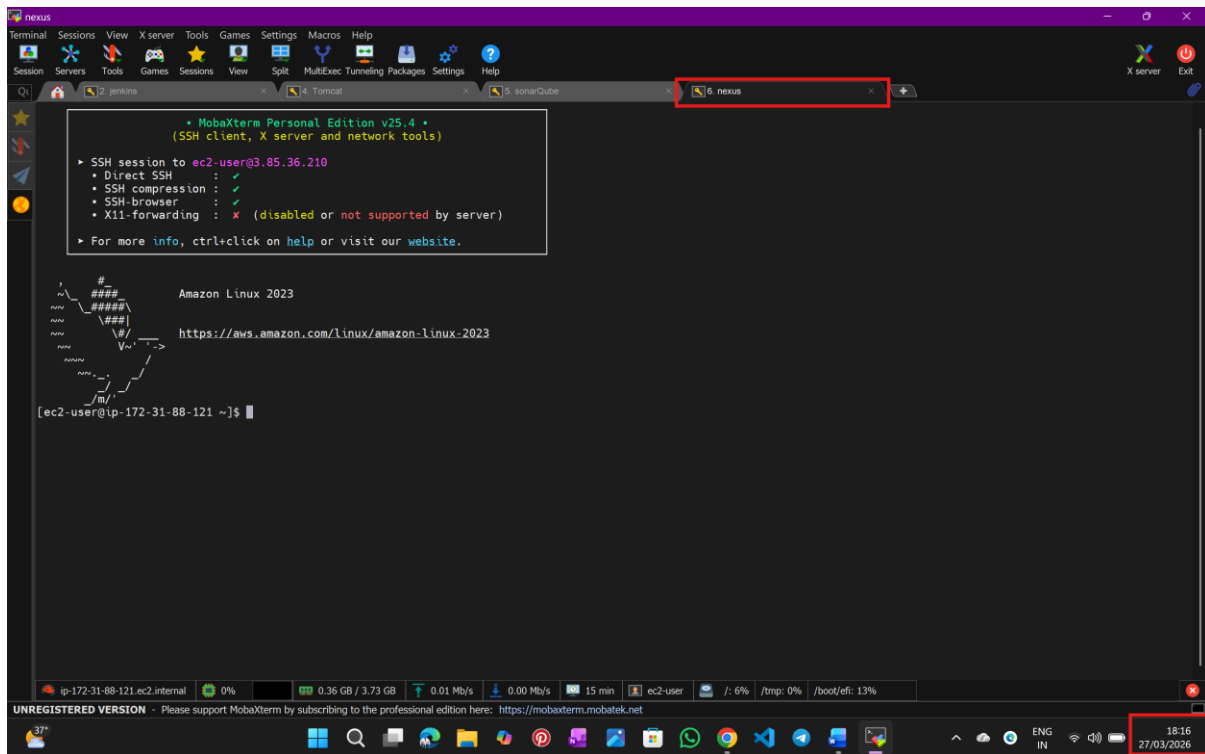
→ using public IP :<port number> access the Jenkins web server

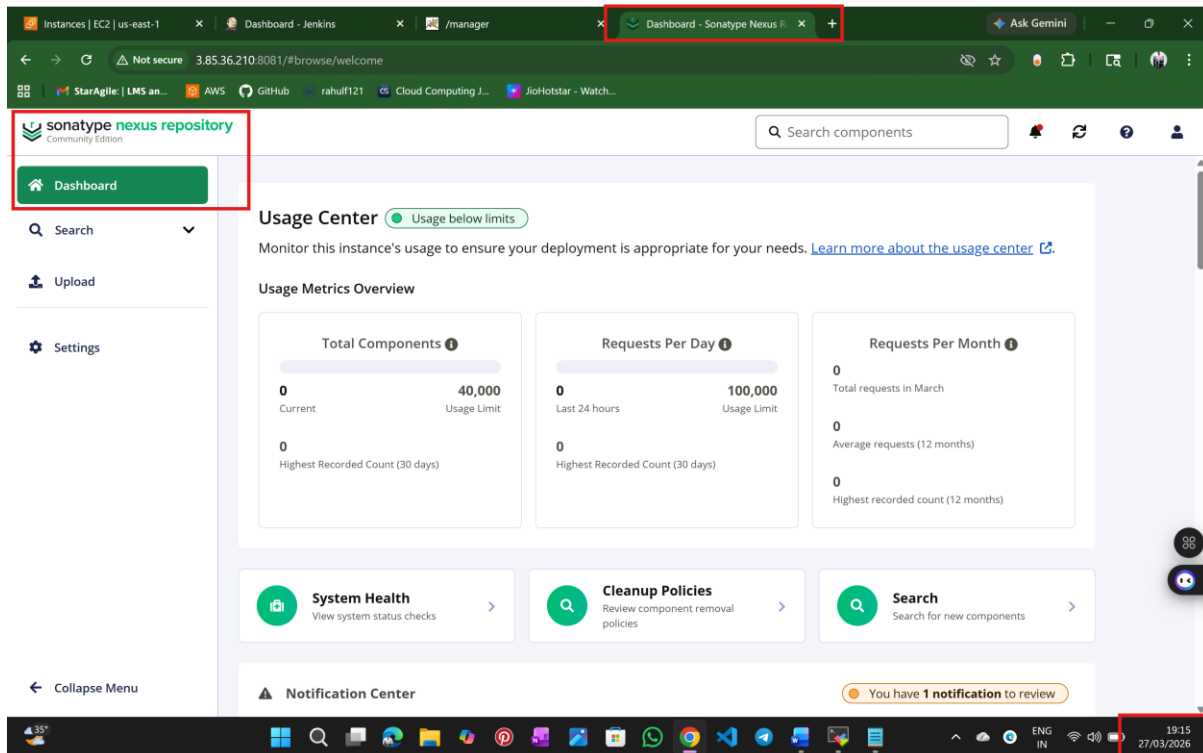
Eg: public IP:8081 → by default Jenkins run on 8081 port number

→ To access the Sonar Qube Dashboard

Username: admin

Password: admin





On Sonar Qube server

→ connect to mobXterm with public IP of the server

→ install docker

→ install maven

→ use the below command

`docker run -d -p <host port no>:<container port no> --name <container name> <docker image name>`

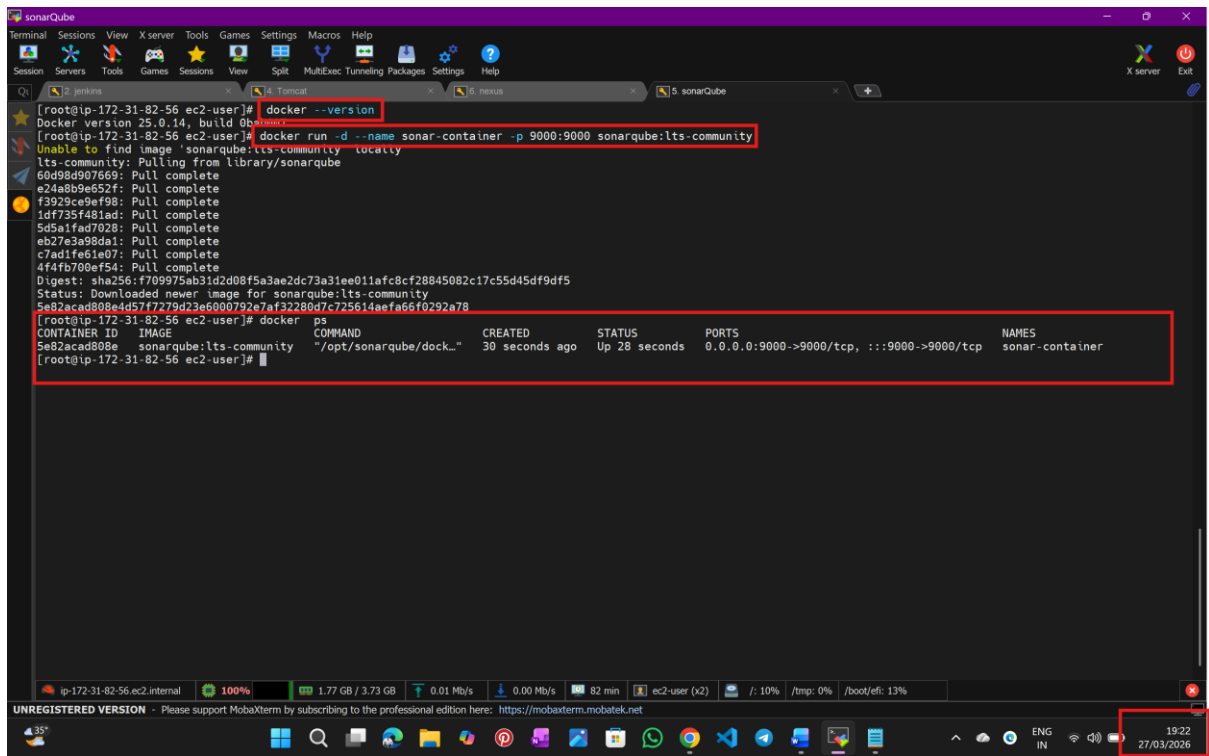
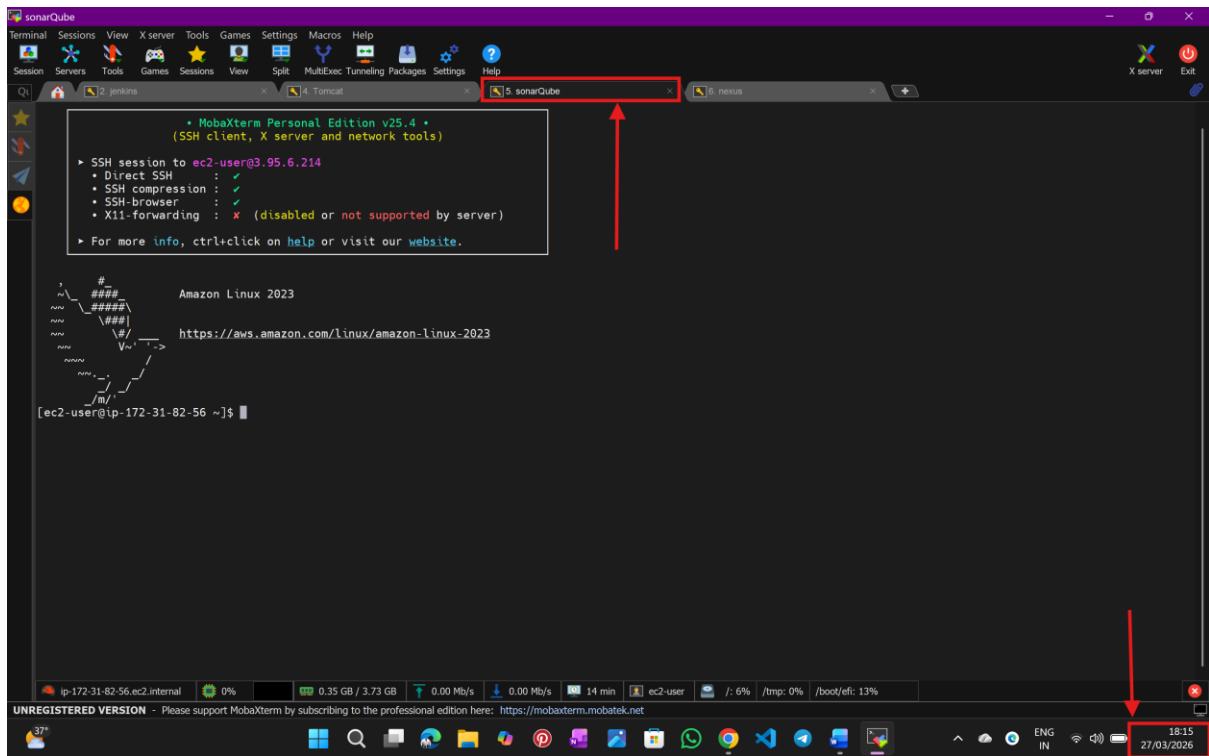
→ using public IP :<port number> access the Jenkins web server

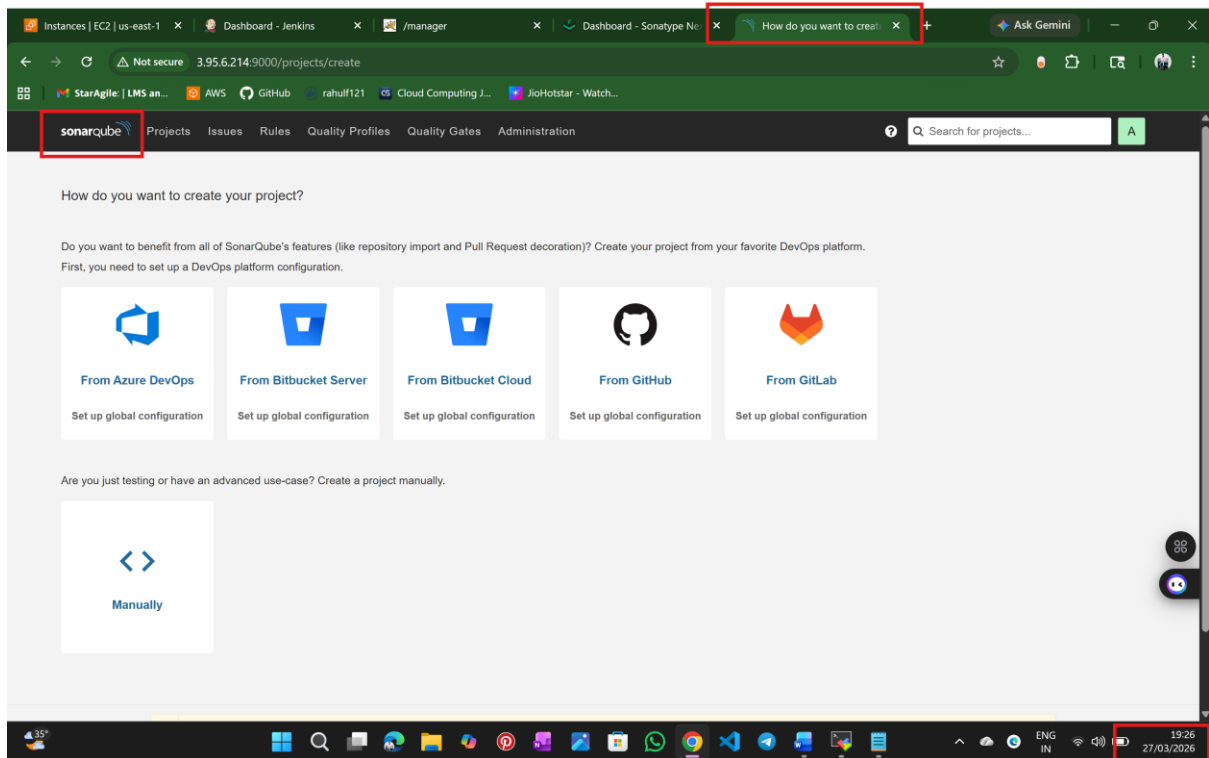
Eg: public IP:9000 → by default Jenkins run on 9000 port number

→ To access the Sonar Qube Dashboard

Username: admin

Password: admin

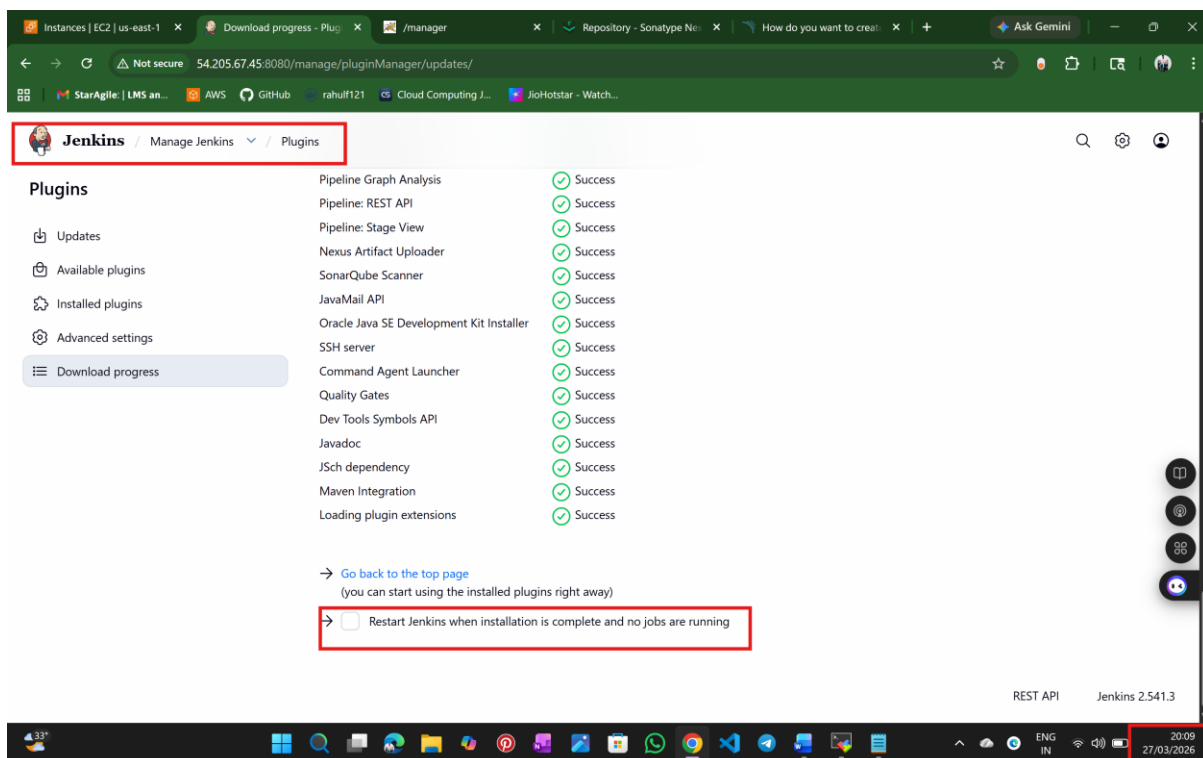




Step 4: Inside Jenkins config the details

1. For Tomcat
 - plugin: deploy to container
 - credentials: tomcat
 - tools: add JDK
 - pipeline syntax
2. Tools
 - java
 - maven
 - sonar Qube scanner installer
3. To store artifacts in s3
 - plugin: s3 publisher
 - system: config s3profile → use IAM role → into <bucketname>
 - pipeline syntax

4. To store artifacts in Nexus
 - plugin: config file provider
 - maven integration
 - credentials: nexus
 - pipeline syntax
5. Code quality analysis: sonar Qube
 - plugin: sonar Qube scanner
 - plugin: quality gates
 - credentials: sonar Qube
 - pipeline syntax
6. To get Email Notification
 - plugin: Email extension plugin
 - Email app password
 - credentials: from email app password
 - system: config email notification



Step: 5 Pipeline script

```
pipeline {
  agent any
  tools{
    jdk 'jdk17'
    maven 'maven3'
  }

  stages {
    stage('git checkout') {
      steps {
        git branch: 'main', url: 'https://github.com/rahulnalathati-21/java-tomcat-webapp.git'
      }
    }

    stage('sonar-quality') {
      steps {
        withSonarQubeEnv('rahul') {
          sh 'mvn sonar:sonar'
        }
      }
    }

    stage('maven build') {
      steps {
        sh 'mvn clean package'
      }
    }

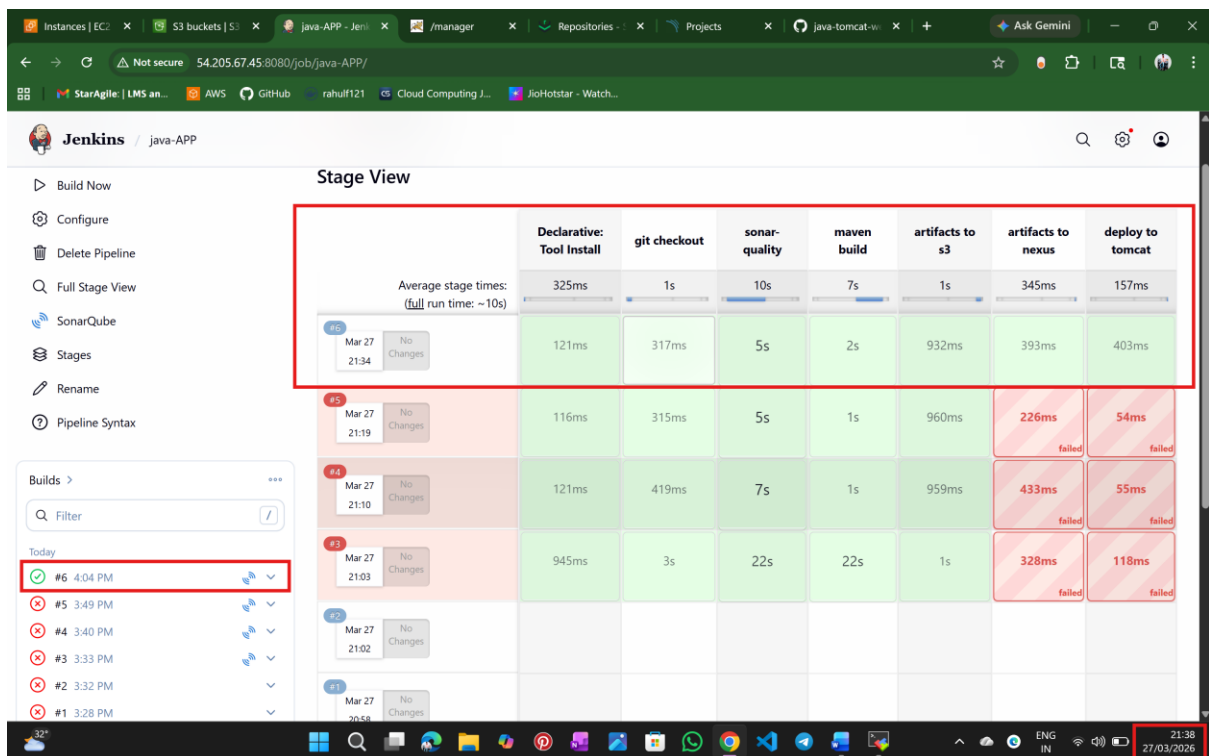
    stage('artifacts to s3') {
      steps {
        sh 'aws s3 cp **/*.war s3://s3-bucket-rahul123/RAHUL-${BUILD_NUMBER}.war --region us-east-1'
      }
    }

    stage('artifacts to nexus') {
      steps {
        nexusArtifactUploader artifacts: [[artifactId: 'java-tomcat-webapp', classifier: '', file: 'target/app.war', type: 'war']],
        credentialsId: 'nexus-creds', groupId: 'com.devops', nexusUri: '3.85.36.210:8081', nexusVersion: 'nexus3', protocol: 'http',
        repository: 'rahul', version: '1.0'
      }
    }

    stage('deploy to tomcat') {
      steps {
        deploy adapters: [tomcat9(alternativeDeploymentContext: '', credentialsId: 'tomcat-creds', path: '', url:
'http://3.83.205.75:8080')], contextPath: 'Java-App', war: '**/*.war'
      }
    }
  }
}
```

Step:6 Result or output

Build now – To deploy into the tomcat server



The java web application successfully deployed in tomcat

Tomcat Web Application Manager

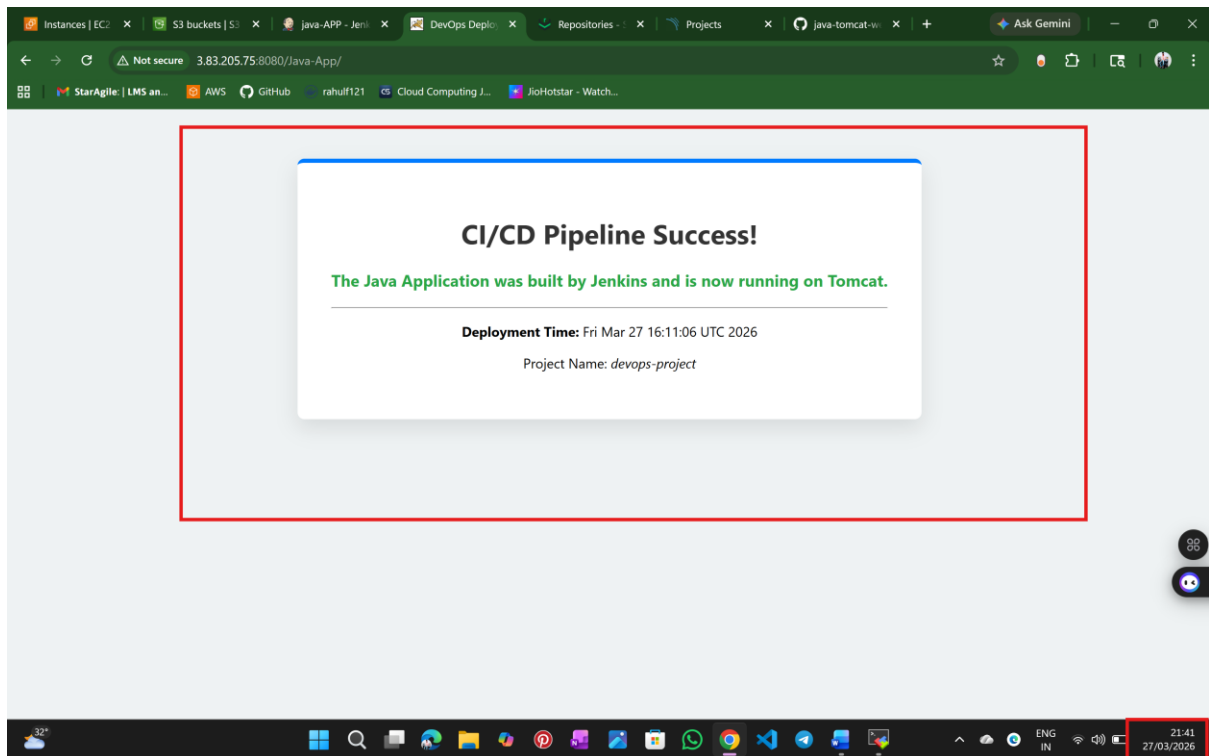
Message: OK

Manager

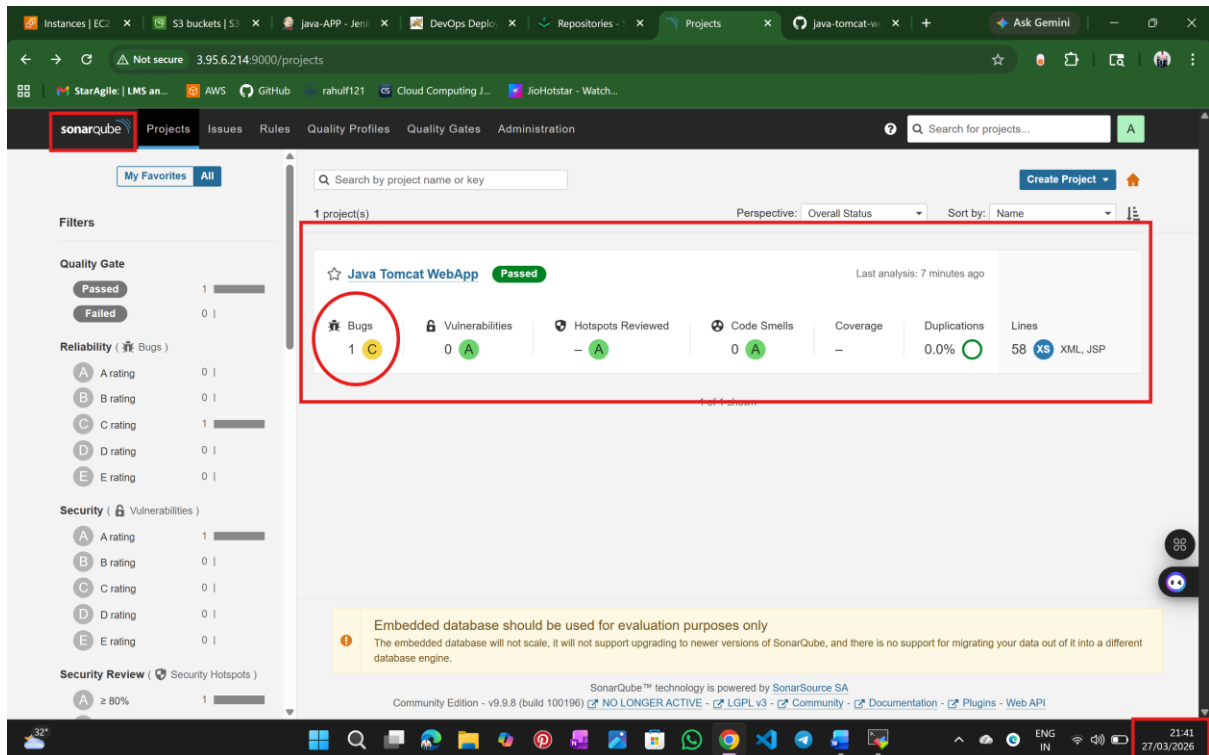
Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy
/Java-App	None specified	DevOps Practice Project	true	0	Start Stop Reload Undeploy
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy

Deploy

Deploy directory or WAR file located on server



The Result of sonarQube



The artifacts stored in Nexus

The screenshot shows the Sonatype Nexus repository interface. The left sidebar contains navigation options: Dashboard, Search, Browse (highlighted), Upload, and Settings. The main content area displays the 'Browse' view for the repository 'raahul'. A tree structure shows the path 'com/devops/java-tomcat-webapp/1.0', with the following artifacts listed: 'java-tomcat-webapp-1.0.war', 'java-tomcat-webapp-1.0.war.md5', and 'java-tomcat-webapp-1.0.war.sha1'. On the right, a 'Summary' panel provides details for the selected component: Repository (raahul), Format (maven2), Group (com.devops), Name (java-tomcat-webapp), Version (1.0), Most popular version, Age, and Popularity. Below the summary, the 'Usage' section shows 'Apache Maven' and a snippet for insertion into a pom.xml file. The browser's address bar shows the URL '3.85.36.210:8081/#browse/browse:raahul:com%2Fdevops%2Fjava-tomcat-webapp%2F1.0'. The system tray at the bottom right shows the time '21:43' and date '27/03/2026'.

The artifacts stored in s3 bucket

The screenshot shows the Amazon S3 console interface. The left sidebar contains navigation options: Buckets, Access management and security, and Storage management and insights. The main content area displays the 's3-bucket-raahul123' bucket. The 'Objects' tab is selected, showing a list of four objects: 'RAHUL-3.war', 'RAHUL-4.war', 'RAHUL-5.war', and 'RAHUL-6.war'. Each object is a 'war' file, 2.2 KB in size, and stored in the 'Standard' storage class. The 'Last modified' dates are: March 27, 2026, 21:04:34 (UTC+05:30), March 27, 2026, 21:10:14 (UTC+05:30), March 27, 2026, 21:19:56 (UTC+05:30), and March 27, 2026, 21:34:18 (UTC+05:30). The browser's address bar shows the URL 'us-east-1.console.aws.amazon.com/s3/buckets/s3-bucket-raahul123?region=us-east-1&tab=objects'. The system tray at the bottom right shows the time '21:44' and date '27/03/2026'.

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